Backend Development

Frontend: HTML, CSS, JS, React.js, Next.js

Backend: Node.js, Express.js, Spring, Spring Boot, Flask, Django, ASP.NET, C#, .NET + Database → Relational, NoSQL

Database → structured collection of data

[DBMS (Database Management System)]

① RDBMS (SQL) → Relational DBMS

Structured Database

Table → rows, columns → Tuples, attributes

Product Table

PID	Name	Price	Rating

e.g. OracleDB, MySQL, MariaDB, PostgreSQL } RDBMS → SQL

What DBMS do?
- Data storage
- Data retrieval, manipulation
- Data security
- Data integrity
- Scalability
- Backup, Recovery

② NOSQL DBMS
- unstructured data is changing rapidly
- ① Document store (JSON like format) → MongoDB
- ② key-value → Redis, DynamoDB
- ③ Column store → Cassandra
- ④ graph Database → AllegroGraph, Neptune, Neo4j

③ OODBMS (object-oriented) → versant

④ Hierarchical DBMS → IMS → Tree data structure

⑤ Network DBMS → IDMS

⑥ Distributed DBMS → DynamoDB, cassandra.

⑦ Vector DB → AI

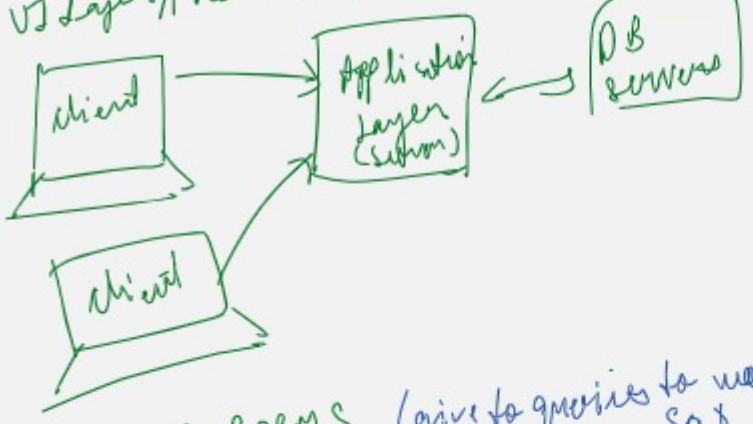
What to do?

Backend / server-side
- ① Server logic (Building the server) — Ch-4
- ② Database Management — Ch-1, 2
- ③ API Development — Ch-4

Backend Developer skills
- ① → DSA
- ② → Prog lang & framework → JS, Node, Express.
- ③ → Database Management systems → MySQL, PostgreSQL, mongodb, oracle DB
- ④ → hosting platform.

3-tier Arch of DBMS

UI Layer / Presentation



Data Language in RDBMS (give to queries to work in RDBMS)

SQL → Structured Query Language

DDL → create, alter, drop, Truncate, rename

DML → insert, update, delete, merge, select

DCL → grant, Revoke

Transaction → commit, Rollback, savepoint

MariaDB → install